

Question_1

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.1      ✓ stringr    1.5.2
## ✓ ggplot2     4.0.0      ✓ tibble     3.3.0
## ✓ lubridate  1.9.4      ✓ tidyr      1.3.1
## ✓ purrr      1.1.0
## — Conflicts — tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()    masks stats::lag()
## ⓘ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(plotly)
```

```
##
## Attaching package: 'plotly'
##
## The following object is masked from 'package:ggplot2':
##
##   last_plot
##
## The following object is masked from 'package:stats':
##
##   filter
##
## The following object is masked from 'package:graphics':
##
##   layout
```

```
library(htmlwidgets)
```

```
## Warning: package 'htmlwidgets' was built under R version 4.5.2
```

```
library(scales)
```

```
##  
## Attaching package: 'scales'  
##  
## The following object is masked from 'package:purrr':  
##  
##   discard  
##  
## The following object is masked from 'package:readr':  
##  
##   col_factor
```

```
library(ggthemes)
```

```
energy <- read_csv(  
  "data/data_raw/owid_energy.csv", show_col_types = FALSE  
)
```

```
## New names:  
## • `` -> `...1`
```

```
co2 <- read_csv(  
  "data/data_raw/owid_co2.csv", show_col_types = FALSE  
)
```

```
## New names:  
## • `` -> `...1`
```

```
energy_sel <- energy |>
  select(
    country, iso_code, year,
    renewables_share_elec,
    low_carbon_share_elec,
    fossil_share_elec,
    renewables_electricity, # TWh
    fossil_electricity      # TWh
  )

co2_sel <- co2 |>
  select(
    country, iso_code, year,
    co2, # total CO2 emissions (Mt)
    co2_per_capita,
    consumption_co2_per_capita,
    gdp,
    population
  )

cr_energy_co2 <- energy_sel |>
  left_join(co2_sel,
    by = c("country", "iso_code", "year")) |>
  filter(country == "Costa Rica",
    year >= 1990, year <= 2023)
```

```
# Get 2010 baseline (if missing, pick nearest year you actually have)
```

```
cr_base_2010 <- cr_energy_co2 |>
```

```
  filter(year == 2010) |>
```

```
  pull(co2_per_capita) |>
```

```
  first()
```

```
cr_pathway <- cr_energy_co2 |>
```

```
  mutate(
```

```
    co2pc_target_15c = case_when(
```

```
      year < 2010 ~ NA_real_,
```

```
      year >= 2010 & year <= 2050 ~
```

```
        cr_base_2010 * pmax(0, 1 - (year - 2010) / (2050 - 2010)),
```

```
      year > 2050 ~ 0
```

```
    )
```

```
  )
```

```

# Trim years to keep only what matters visually
cr_trim <- cr_pathway |>
  filter(year >= 2000)

max_y <- max(cr_trim$co2_per_capita, na.rm = TRUE)

p1 <- ggplot(cr_trim, aes(x = year)) +

  # Light shaded window only for evaluation period
  annotate(
    "rect", xmin = 2010, xmax = 2023,
    ymin = -Inf, ymax = Inf,
    fill = "grey90", alpha = 0.5
  ) +

  # 1.5°C pathway
  geom_line(
    aes(y = co2pc_target_15c),
    colour = "#D55E00",
    linewidth = 1.1,
    linetype = "dashed"
  ) +

  # Observed CO2 per capita
  geom_line(
    aes(y = co2_per_capita),
    colour = "#1A1A1A",
    linewidth = 1.4
  ) +

  # Annotation
  annotate(
    "text",
    x = 2015, y = max_y * 0.65,
    label = "2023 gap: +0.55 t/person\n(~50% above pathway)",
    hjust = 0, size = 4,
    colour = "red"
  ) +

```

```

geom_hline(yintercept = 0, linetype = "dotted", colour = "grey60") +
geom_vline(xintercept = 2010, linetype = "dashed", colour = "grey50") +
annotate(
  "text", x = 2010, y = 0.15,
  label = "2010 baseline",
  angle = 90, vjust = -0.3, hjust = 0,
  size = 3, colour = "grey40"
)+

labs(
  title = expression(
    "Costa Rica's " * CO[2] * " per capita vs a 1.5" * degree * "C-compatible pathway"),
  subtitle = "Electricity is already low-carbon; reductions now depend on transport and other fossil uses",
  x = "Year",
  y = expression("CO"[2] * " emissions per capita (tonnes/person)"),
  caption = expression("Source: OWID " * CO[2] * " and Energy datasets.")
) +

coord_cartesian(ylim = c(0, max_y * 1.05)) +

theme_clean(base_size = 14) +
theme(
  legend.position = "none",
  panel.grid.minor = element_blank(),
  panel.grid.major = element_line(colour = "grey90"),
  plot.title = element_text(face = "bold"),
  plot.subtitle = element_text(size = 11, colour = "#555555")
)

```

```

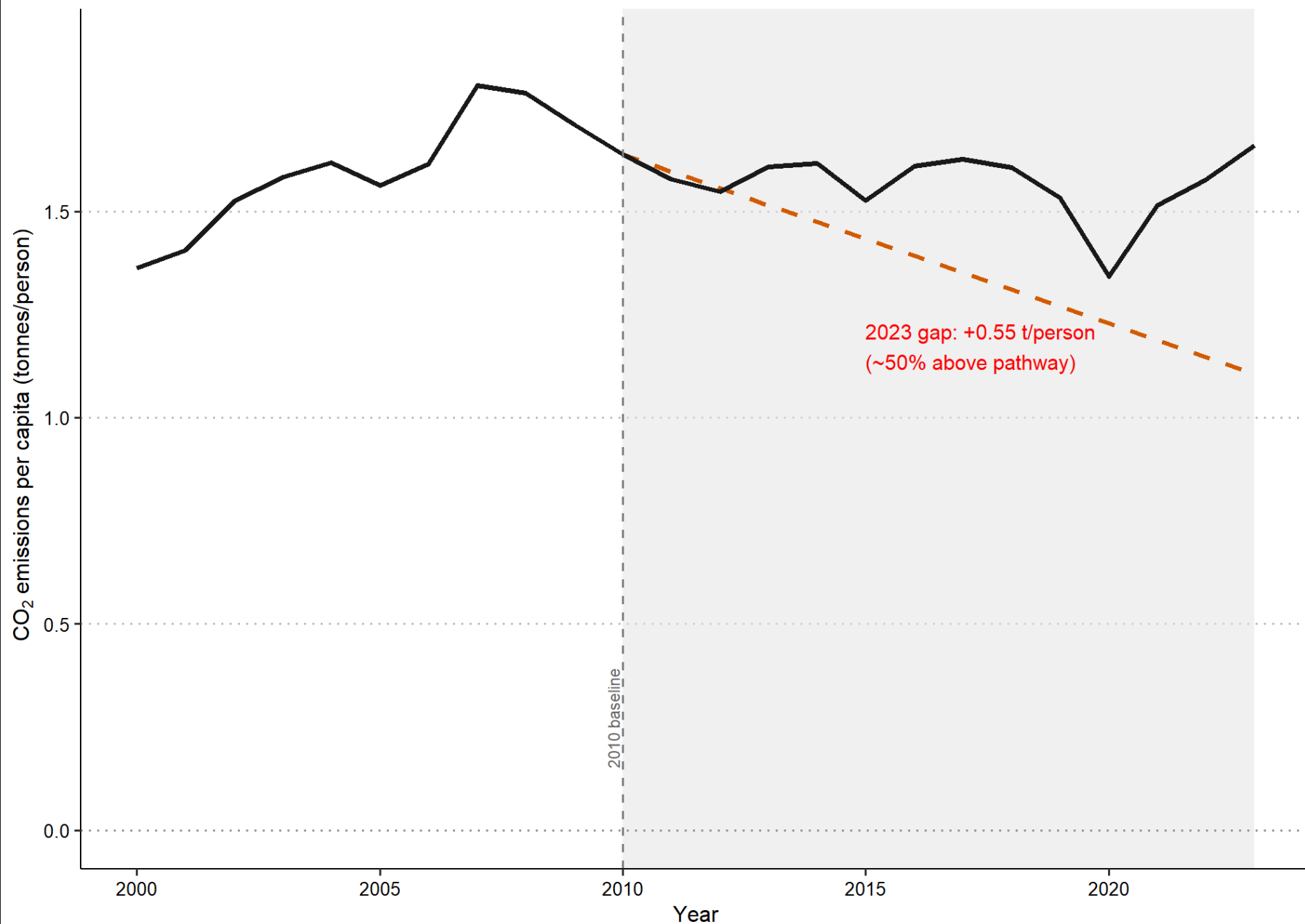
## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
## i Please use the `linewidth` argument instead.
## i The deprecated feature was likely used in the ggthemes package.
## Please report the issue at <https://github.com/jrnold/ggthemes/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## (`geom_line()`).
```

Costa Rica's CO₂ per capita vs a 1.5°C-compatible pathway

Electricity is already low-carbon; reductions now depend on transport and other fossil uses



Source: OWID CO₂ and Energy datasets.